Eloi MARTINET

Current situation

2024 - **Post-doc in Variational methods in Machine Learning**, *JMU*, Würzburg, Germany.

Publications and preprints

2026 **On the (non-)closedness of the set of ReLU networks**, *with Leon Bungert*, Preprint.

2026 **Numerical exploration of the range of shape functionals using neural networks**, *with Ilias Ftouhi*, Preprint.

2026 **Parametrizing Convex Sets Using a Sublinear Layer**, Transactions of Machine Learning Research.

2025 **Spherical caps do not always maximize Neumann eigenvalues on the sphere**, *with Dorin Bucur, Richard S. Laugesen and Mickaël Nahon*, Geometric and Functional Analysis.

2025 **Meshless Shape Optimization using Neural Networks and Partial Differential Equations on Graphs**, *with Leon Bungert*, Scale Space and Variational Methods in Computer Vision.

2024 **Sharp inequalities for Neumann eigenvalues on the sphere**, *with Dorin Bucur and Mickael Nahon*, Journal of Differential Geometry.

2024 **Numerical optimization of Neumann eigenvalues on the sphere**, Journal of Computational Physics.

2023 **Maximization of Neumann Eigenvalues**, *with Dorin Bucur and Edouard Oudet*, Archive for Rational Mechanics and Analysis.

Conferences, workshops and seminars

2026 **GT CalVa Seminar**, Paris.

2026 **Workshop MATHSIA**, Bordeaux.

2026 **FBP Conference**, Berlin.

2026 **AIMS Conference**, Athens.

2026 **Shape Optimization, Geometric Inequalities and Related Topics**, *Centro Congressi Federico II*, Naples.

2025 **Meeting of the ANR Stoiques**, *Sorbonne*, Paris.

2025 **Mathematics of Machine Learning 2025**, Hamburg, (poster session).

2025 **ENUMATH 2025**, Heidelberg.

2025 **CalcVarNL 2025**, Roermond.

2025 **Synergies of Machine Learning and Numerics**, Osaka.

2024 **MIPA Seminar**, *MIPA*, Nîmes.

2024 **Seminar Mathematics of Machine Learning and Applied Analysis**, *JMU*, Würzburg.

2024 **Partial Differential Equations seminar**, *IECL*, Nancy.

2023 **POEMS seminar**, *ENSTA*, Paris.

2023 **Calculus of Variations and Applications**, *Univesité Paris Cité*, Paris.

2023 **Partial Differential Equations seminar**, *IRMA*, Strasbourg.

2023 **Shape Optimization, Geometric Inequalities and Related Topics**, *Dip. Mat. Appl.*, Naples.

2021 **Meeting of the ANR SHAPO**, *Autrans*.

2019 **EDPs2 discussion group**, *USMB*, Chambery.

Research Stays

2026 **One-week invitation from Charles Dapogny**, *Paris*, Germany.

2025 **One-week invitation from Martin Rumpf**, *Bonn*, Germany.

Teaching

2026-2027 **Seminar of Regularization in Machine Learning**, *JMU*, Würzburg.

2026-2027 **PDEs on Graphs**, *JMU*, Würzburg.

2026 **Lectures on Finite Elements Methods and Scientific ML**, *JMU*, Würzburg.

2025-2026 **PDEs on Graphs**, *JMU*, Würzburg.

2025 **Project of Machine Learning**, *JMU*, Würzburg.

2025 **Lectures on Finite Elements Methods and Physics Informed Neural Networks**, *JMU*, Würzburg.

2024-2025 **Lectures and exercises on Introduction to Data Science**, *JMU*, Würzburg.

2024-2025 **Seminar of Introduction to Machine Learning**, *JMU*, Würzburg.

2024 **Lectures on Finite Elements Methods and Physics Informed Neural Networks**, *JMU*, Würzburg.

2024 **Seminar of Machine Learning on graphs**, *JMU*, Würzburg.

2022-2023 **Tutoring of practical sessions of Fourier Transform and Numerical Analysis**, *ENSIMAG*, Grenoble.

2021-2022 **Tutoring of practical sessions and courses of Basic Analysis**, *UGA*, Grenoble.

2019-2020 **Tutoring of practical sessions of Numerical Analysis**, *USMB*, Chambery.

Supervision

2025 **Supervision of Jana Bardoz, student assistant**, *JMU Würzburg student*.

2025 **Co-supervision of the MSc Internship of Nick Burk**, *JMU Würzburg student*, with Leon Bungert.

2024 **Co-supervision of the MSc Internship of Nicolas Roblet**, *ENSIMAG student*, with Romain Joly.

Outreach

2025 **"Machine Learning on Graphs"**, *one week project for high schoolers*, JMU, Würzburg, Germany.

2025 **"Comment les machines apprennent-elles ?"**, *for teachers*, Collège Maurice Clavel, Avallon, France.

2025 **"Maths et Machines"**, *for middle and high schoolers*, Collège Maurice Clavel and Lycée des Chaumes, Avallon, France.

2023 **"What is it to be a PhD student ?"**, *for high schoolers*, LJK, Grenoble, France.

Administrative responsibilities

2026 **Co-organizer of a Minisymposium at the FBP 2026 Conference**, *Berlin*, (with Matthew Thorpe).

2024- **Organization of the team seminar of Mathematics of Machine Learning and Applied Analysis**, *JMU Würzburg*.

2024- **Reviewer: SSVM, JMLR, ANR project and other.**, *JMU Würzburg*.

Grants

2024 **JMU Seed Grant**, *JMU Würzburg*.
11.000€

Education

2019-2023 **PhD in spectral shape optimization**, *LAMA*, Chambery, Under the supervision of D. BUCUR and E. OUDET.

2020-2021 **Master's degree "Préparation à l'agrégation"**, *UGA*, Grenoble, Ranked 88.

2016-2019 **Master's degree in computer science and applied mathematics**, *ENSIMAG*, Grenoble.
Double Engineer/Master degree programme.

2013–2016 **Classes préparatoires MP***, *Lycée Carnot*, Dijon.
Post secondary academic course specialised in mathematics and physics.

Internships

March-July 2019 **Intership in shape optimization**, *LAMA/LJK*, Chambery/Grenoble.

May-July 2018 **Intership in machine learning and image processing**, *GIPSA-Lab*, Grenoble.